

Solving the Portfolio Optimization Problem

IMW 2026 – Optimization Task

Portfolio optimization theory helps investors determine how to allocate investments among different assets to maximize expected return while minimizing investment risk. Therefore, portfolio optimization can be formulated as a multiobjective optimization problem. In this task, you will solve portfolio optimization problems using both single-objective and multiobjective optimization methods..

Instructions

1. Select 10 Indonesian stocks based on the letters in your name. Example: ADHE KANIA → ANTM, DEWA, HRUM, EXCL, KLBF, ASII, NISP, INDF, AKRA, ACES. You may use a different valid mapping based on your own name.
2. Collect daily closing price data for all selected stocks for the most recent 7-year period. Present the data graphically using line charts (one combined chart).
3. Construct an equally weighted portfolio by investing 10% of capital in each stock. Calculate expected portfolio return and portfolio risk for this portfolio.
4. Apply a metaheuristic optimization method (e.g., Genetic Algorithm, Particle Swarm Optimization, or Differential Evolution) to minimize portfolio risk (ignore the return objective in this scenario). You may use the pymoo library to solve the problem. Tune the algorithm-specific parameters to obtain the best possible solution. Report the optimal portfolio weights, the expected portfolio return, and the portfolio risk obtained.
5. Implement NSGA-II to simultaneously minimize portfolio risk and maximize expected return. Generate and display the Pareto-optimal frontier.
6. Add a minimum-buy threshold constraint: any selected stock must have a portfolio weight of at least 10% (not all stocks need to be selected). Re-run NSGA-II and plot the new Pareto frontier. Compare with the unconstrained case.
7. Add an additional constraint that the maximum number of stocks held in the portfolio is 7. Re-run NSGA-II and plot the Pareto frontier. Compare results with previous scenarios.